

COSM
Unit-2

Expectation and discrete distributions: -

Mean of a Random Variable

Variance and covariance of Random variables

Means and variances of linear combinations of Random variables

Discrete Probability distributions: -

1. Binomial 2. Poisson 3. Geometric.

Expectation: - $E(x) = \sum_{i=1}^n x_i P(x_i)$ where $x_1, x_2, \dots, x_n$ are values taken by x , $P(x_1), P(x_2), \dots, P(x_n)$ are their probabilities.

→ It represents the average value of the random variable.

Properties: - 1. $E(c) = c$ where c is a constant.
2. $E(ax) = aE(x)$ 3. $E(x+y) = E(x) + E(y)$.

Mean of a discrete random variable: - If x takes $x_1, x_2, \dots, x_n$ with probabilities $p_1, p_2, \dots, p_n$ then the mean of $x = \mu = E(x) = \sum_{i=1}^n x_i p_i$.

Variance of a random variable: - It measures the spread of values around the mean.

$$\sigma^2 = \text{Var}(x) = E(x^2) - (E(x))^2 = \sum_{i=1}^n x_i^2 p_i - \mu^2$$

Covariance:— Covariance of x and y is defined as $\text{cov}(x, y) = E(xy) - E(x) \cdot E(y)$.

Significance:— 1. $\text{cov}(x, y) > 0$: x & y increase or decrease together.

2. $\text{cov}(x, y) < 0$: one increases, the other decreases.

3. $\text{cov}(x, y) = 0$: x and y are independent.

Linear combinations:—

Mean of $z = ax + by + c$ is $E(z) = aE(x) + bE(y) + c$.

Variance of $z = ax + by + c$ is $\text{var}(z) = a^2 \text{var}(x) + b^2 \text{var}(y) + 2ab \text{cov}(x, y)$.

→ If x and y are independent then $\text{cov}(x, y) = 0$

$$\therefore \text{var}(z) = a^2 \text{var}(x) + b^2 \text{var}(y).$$

Q. If $E(x) = 2$, $\text{var}(x) = 3$ find $E(3x+4)$ & $\text{var}(3x+4)$

Sol:— $E(3x+4) = 3 \cdot E(x) + 4 = 3(2) + 4 = 10$.

$$\text{var}(3x+4) = 3^2 \text{var}(x) = 9(3) = 27.$$

Q. If $E(x) = 1$, $\text{var}(x) = 2$, $E(y) = 2$, $\text{var}(y) = 1$, x, y are independent, find mean and variance of $z = 2x - y$.

Sol:— x, y are independent, so $\text{cov}(x, y) = 0$.

$$E(2x - y) = 2E(x) - E(y) = 2(1) - 2 = 0.$$

$$\begin{aligned} \text{Var}(2x - y) &= 2^2 \text{var}(x) + 1^2 \text{var}(y) - 2(2)(1) \text{cov}(x, y) \\ &= 4(2) + 1(1) - 4(0) = 9. \end{aligned}$$

Q. If $\text{var}(x) = 4$, $\text{var}(y) = 9$, $\text{cov}(x, y) = 2$ find $\text{var}(x+y)$

$$\begin{aligned} \text{Sol: } \text{var}(x+y) &= 1^2 \text{var}(x) + 1^2 \text{var}(y) + 2(1)(1) \text{cov}(x, y) \\ &= 4 + 9 + 2(2) = 17. \end{aligned}$$

Bernoulli trial:- It is an experiment with only 2 outcomes. 1. Success (S), 2. Failure (F).

If probability of success = p , then the probability of failure = $q = 1 - p$.

Binomial distribution:- It is the probability distribution of the no. of successes in n independent Bernoulli trials, where each trial has the same probability of success ' p '.

Binomial Probability mass function:- It gives the probability of getting exactly ' r ' successes in n independent Bernoulli trials.

If x is the no. of successes in n trials, then it takes the values $0, 1, 2, 3, \dots, n$.

$$P(X=r) = \binom{n}{r} p^r q^{n-r}$$

Q- Describe the mean and variance of a Binomial Distribution.

Let the probability distribution of x , the no. of successes in n independent trials where probability of success is p and the probability of failure is q , is $P(X=r) = \binom{n}{r} p^r q^{n-r}$.

$$\text{and mean of } x = \mu = \sum_{r=0}^n x_i P(X=x_i)$$

$$= x_0 P(X=x_0) + x_1 P(X=x_1) + \dots + x_n P(X=x_n)$$

$$\begin{aligned}
&= 1 \cdot p(x=1) + 2 \cdot p(x=2) + 3 \cdot p(x=3) \dots + n \cdot p(x=n) \\
&= 1 \cdot \binom{n}{1} p^1 q^{n-1} + 2 \cdot \binom{n}{2} p^2 q^{n-2} + 3 \cdot \binom{n}{3} p^3 q^{n-3} \dots + n \binom{n}{n} p^n q^0 \\
&= npq^{n-1} + 2 \frac{n(n-1)}{2!} p^2 q^{n-2} + 3 \frac{n(n-1)(n-2)}{3!} p^3 q^{n-3} \dots + n p^n \\
&= np [q^{n-1} + (n-1) p q^{n-2} + \frac{(n-1)(n-2)}{2!} p^2 q^{n-3} \dots + p^{n-1}] \\
&= np [p+q]^{n-1} = np(1) = np \quad (\because p+q=1)
\end{aligned}$$

$$\therefore \boxed{\text{Mean} = \mu = np}$$

$$\text{Variance}(x) = \sigma^2 = \sum_{i=0}^n x_i^2 p(x=x_i) - \mu^2$$

$$\begin{aligned}
&= x_0^2 p(x=x_0) + x_1^2 p(x=x_1) \dots \\
&= \sum_{r=0}^n r^2 p(x=r) - \mu^2 = \sum_{r=0}^n (r(r-1) + r) p(x=r) - \mu^2 \\
&= \sum_{r=0}^n r(r-1) p(x=r) + \sum_{r=0}^n r p(x=r) - \mu^2 \\
&= 2 \cdot 1 \cdot p(x=2) + 3 \cdot 2 \cdot p(x=3) \dots + n(n-1) p(x=n) + \mu - \mu^2 \\
&= 2 \cdot 1 \cdot \binom{n}{2} p^2 q^{n-2} + 3 \cdot 2 \cdot \binom{n}{3} p^3 q^{n-3} \dots + n(n-1) \binom{n}{n} p^n q^0 + \mu - \mu^2 \\
&= 2 \cdot 1 \cdot \frac{n(n-1)}{2!} p^2 q^{n-2} + 3 \cdot 2 \cdot \frac{n(n-1)(n-2)}{3!} p^3 q^{n-3} \dots + n(n-1) p^n + \mu - \mu^2 \\
&= n(n-1) p^2 (q^{n-2} + (n-2) p q^{n-3} \dots + p^{n-2}) + np - n^2 p^2 \\
&= n(n-1) p^2 [p+q]^{n-2} + np - n^2 p^2 \\
&= n(n-1) p^2 (1) + np - n^2 p^2 \quad (\because p+q=1) \\
&= n^2 p^2 - np^2 + np - n^2 p^2 = np(1-p) = npq \\
&\therefore \boxed{\text{Variance}(x) = \sigma^2 = npq}
\end{aligned}$$

Recurrence relation for the Binomial distribution:-

$$p(r+1) = \frac{(n-r)}{(r+1)} \frac{p}{q} \cdot p(r)$$

Q. Two dice are thrown 5 times. Find the probability of getting 7 as sum (i) atleast once (ii) 2 times
(iii) $P(1 < x < 5)$.

Sol Let x be the no. of times getting 7 as sum when 2 dice are thrown 5 times.

Then x takes 0, 1, 2, 3, 4, 5.

p = The probability of getting sum 7 in a single throw of a pair of dice = $\frac{6}{36} = \frac{1}{6}$

Then $q = 1 - \frac{1}{6} = \frac{5}{6}$.

No. of trials = 5.

$$(i) \quad P(\text{atleast once}) = P(x \geq 1) = 1 - P(x=0) = 1 - \binom{5}{0} p^0 q^5 = 1 - q^5 = 1 - \left(\frac{5}{6}\right)^5 = 1 - \frac{3125}{7776} = 1 - 0.4018 = 0.598$$

$$(ii) \quad P(2 \text{ times}) = P(x=2) = \binom{5}{2} p^2 q^3 = \binom{5}{2} \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^3 = 0.16$$

$$(iii) \quad P(1 < x < 5) = P(x=2) + P(x=3) + P(x=4)$$

$$= \binom{5}{2} p^2 q^3 + \binom{5}{3} p^3 q^2 + \binom{5}{4} p^4 q^1$$

$$= \binom{5}{2} \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^3 + \binom{5}{3} \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^2 + \binom{5}{4} \left(\frac{1}{6}\right)^4 \left(\frac{5}{6}\right)^1$$

Q. If the probability of a defective bolt is $\frac{1}{8}$.

Find (i) The mean (ii) The variance for the distribution of defective bolts of 640.

Sol here $n=640$, $p=\frac{1}{8} \rightarrow q=1-p=\frac{7}{8}$.

$$(i) \quad \text{mean} = np = \frac{640}{8} = 80$$

$$(ii) \quad \text{variance} = npq = 640 \times \frac{1}{8} \times \frac{7}{8} = 70$$

Q. Out of 800 families with 5 children each, how many would you expect to have (a) 3 boys (b) 5 girls (c) either 2 or 3 boys (d) at least one boy. Assume equal probabilities for boys and girls.

Sol - no. of families = 800. no. of children in each family = 5.

Let x be the no. of boys in each family.

$$p = \text{probability of each boy} = \frac{1}{2}$$

$$q = \frac{1}{2}$$

$$(a) \cdot P(x=3) = {}^5C_3 p^3 q^2 = {}^5C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^2 = \frac{5}{16},$$

per family.

$\therefore$ For all 800 families, $\frac{5}{16} \times 800 = 250$ families have 3 boys.

(b) 5 girls.

$$P(x=0) = {}^5C_0 p^0 q^5 = \left(\frac{1}{2}\right)^5 = \frac{1}{32}$$

$\therefore$ for all 800 families, $\frac{1}{32} \times 800 = 25$ families have 5 girls.

(c) either 2 or 3 boys.

$$= P(x=2) + P(x=3) = \frac{5}{8} \text{ per family.}$$

$$\text{Total} = \frac{5}{8} \times 800 = 500 \text{ families.}$$

$$(d) P(\text{at least one boy}) = P(x \geq 1)$$

$$= 1 - P(x=0) = 1 - {}^5C_0 p^0 q^5 = 1 - \frac{1}{2^5} = 1 - \frac{1}{32}$$

$$= \frac{31}{32}$$

$\therefore$ Expected no. of families with at least one boy = $\frac{31}{32} \times 800 = 775$.

Q. The probability that John hits a target is $\frac{1}{2}$. He fires 6 times. Find the Probability that he hits the target (i) exactly 2 times (ii) more than 4 times (iii). at least once.

Sol:- No. of trials = 6, probability of hitting the target = $p = \frac{1}{2} \rightarrow q = 1 - p = 1 - \frac{1}{2} = \frac{1}{2}$.

(i) $P(X=2) = 0.2344$ (ii) $P(X > 4) = P(X=5) + P(X=6)$
 $= 0.1094$ (iii). $P(X \geq 1) = 1 - P(X=0) = 0.9844$

Q. The mean of Binomial Distribution is 3 and the variance is $\frac{9}{4}$. Find (i) the value of n (ii) $P(X \geq 7)$ (iii) $P(1 \leq X \leq 6)$.

Sol:- Given $np = 3$ and $npq = \frac{9}{4}$.

(i) $\frac{npq}{np} = \frac{9/4}{3} = \frac{9}{4} \times \frac{1}{3} = \frac{3}{4} \therefore q = \frac{3}{4} \rightarrow p = 1 - \frac{3}{4} = \frac{1}{4}$ and $np = 3 \Rightarrow n(\frac{1}{4}) = 3 \Rightarrow n = 12$.

(ii) $P(X \geq 7) = P(X=7) + P(X=8) + P(X=9) + P(X=10) + P(X=11) + P(X=12)$

$= \binom{12}{7} p^7 q^5 + \binom{12}{8} p^4 q^8 = 0.1446$.

(iii). $P(1 \leq X < 6) = P(X=1) + \dots + P(X=5) = 0.82$.

Q. Find the probability of getting an even no. 3 or 4 or 5 times in throwing ^{times} 10 dice. Using Binomial distribution.

Sol:- Let x be the no. of ^{times} getting an even no. in a throw of a die in 10 times.

Then $p = \text{probability of success} = \frac{1}{2}$
 $q = \text{probability of failure} = \frac{1}{2}$
 $n = 10$

$$\therefore P(X=3) = \binom{10}{3} \left(\frac{1}{2}\right)^7 \left(\frac{1}{2}\right)^3 = 0.112$$

$$P(X=4) = \binom{10}{4} \left(\frac{1}{2}\right)^6 \left(\frac{1}{2}\right)^4 = 0.2$$

$$P(X=5) = \binom{10}{5} \left(\frac{1}{2}\right)^5 \left(\frac{1}{2}\right)^5 = 0.246$$

Q. A fair coin is tossed 6 times. Find the probability of getting 4 heads.

Sol The no. of times that a fair coin is tossed = $n = 6$.

Let X be the no. of times getting 4 heads in 6 trials.

then X takes 0, 1, 2, 3, 4.

p = probability of success = $\frac{1}{2}$

q = probability of failure = $\frac{1}{2}$

$$\therefore P(X=4) = \binom{6}{4} \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^2 = 0.2344$$

Q. Determine the probability of getting the sum 6 exactly 3 times in 7 throws with a pair of fair dice.

Sol No. of times that 2 dice are thrown = $T = n$

Let X be the no. of times getting 6 as sum on 2 dice in 7 trials.

then X takes 0, 1, 2, 3, 4, 5, 6, 7.

p = probability of success = $\frac{5}{36}$

q = probability of failure = $\frac{31}{36}$

probability of getting sum 6 exactly
3 times = $P(X=3) = \binom{5}{3} \left(\frac{5}{36}\right)^3 \left(\frac{31}{36}\right)^4 = 0.0516$.

Q. Fit a binomial distribution to the following data

x	0	1	2	3	4	5
f	2	14	20	34	22	8

Sol here $n=5$, $N = \sum f_i = 2+14+20+34+22+8 = 100$.

$$\text{Mean} = \mu = \frac{\sum f_i x_i}{\sum f_i} = \frac{0+14+40+102+88+40}{100} = \frac{284}{100} = 2.84$$

$\therefore$ Mean of the binomial distribution,

$$np = 2.84.$$

$$\rightarrow (5)p = 2.84 \rightarrow p = \frac{2.84}{5} = 0.568$$

$$\text{and } q = 1-p = 0.432$$

hence the Binomial distribution to be fitted

$$\text{is } N(p+q)^n = 100(0.568 + 0.432)^5$$

$$= 100 \left(\binom{5}{0} (0.568)^0 (0.432)^5 + \binom{5}{1} (0.568)^1 (0.432)^4 \right.$$

$$+ \binom{5}{2} (0.568)^2 (0.432)^3 + \binom{5}{3} (0.568)^3 (0.432)^2 +$$

$$\left. \binom{5}{4} (0.568)^4 (0.432)^1 + \binom{5}{5} (0.568)^5 (0.432)^0 \right)$$

$$= 100 [0.015 + 0.0989 + 0.260 + 0.341 + 0.224 + 0.059]$$

$$= 1.5 + 9.89 + 26 + 34.1 + 22.4 + 5.9 \quad \text{Theoretical frequencies}$$

x	0	1	2	3	4	5
f	2	14	20	34	22	8
$E(x)$	2	10	26	34	22	6

since the frequencies are always integers, we take the nearest integers.

Q. 7 coins are tossed and the no. of heads are noted. The experiment is repeated 128 times and the following distribution is obtained.

No. of heads	0	1	2	3	4	5	6	7	Total
frequency	7	6	19	35	30	23	7	1	128

Fit a binomial distribution assuming

- (a) the coin is unbiased (b) the even nature of the coin is not known.

Sol (a) the coin is unbiased.

$$n=7, p=\frac{1}{2}, q=\frac{1}{2}$$

$$P(x=0) = \frac{1}{2^7}$$

$$P(x=1) = 7/2^7$$

$$P(x=2) = 21/2^7$$

$$P(x=3) = 35/2^7$$

$$P(x=4) = 35/2^7$$

$$P(x=5) = 21/2^7$$

$$P(x=6) = 7/2^7$$

$$P(x=7) = 1/2^7$$

$$128 P(x=0) = 1$$

$$128 P(x=1) = 7$$

$$128 P(x=2) = 21$$

$$128 P(x=3) = 35$$

$$128 P(x=4) = 35$$

$$128 P(x=5) = 21$$

$$128 P(x=6) = 7$$

$$128 P(x=7) = 1$$

~~$$128 P(x=0) = 1$$~~

~~$$128 P(x=1) = 7$$~~

These are the theoretical or Expected frequency $n \cdot P(x)$.

X	0	1	2	3	4	5	6	7
f	7	6	19	35	30	23	7	1
$\frac{T(f)}{f}$	1	7	21	35	35	21	7	1

(b) The nature of the coin is not known.

$$\mu = \frac{\sum fixi}{\sum fi} = \frac{433}{128} = 3.383.$$

$$\therefore np = 3.383.$$

$$7p = 3.383$$

$$\therefore p = \frac{3.383}{7} = 0.4833.$$

$$\text{and } q = 1 - p = 1 - 0.4833 = 0.5167$$

$$P(x=0) = \binom{7}{0} (0.4833)^0 (0.5167)^7 \quad \left[\begin{array}{l} 28 \\ 128 \end{array} \right] \quad \begin{array}{l} p(x=0) = 1.25 \\ p(x=1) = 8.209 \end{array} = 1$$

$$P(x=1) = \binom{7}{1} (0.4833)^1 (0.5167)^6 \quad \left[\begin{array}{l} 28 \\ 128 \end{array} \right] \quad \begin{array}{l} p(x=1) = 8.209 \\ p(x=2) = 23.0144 \end{array} = 8$$

$$P(x=2) = \binom{7}{2} (0.4833)^2 (0.5167)^5 \quad \left[\begin{array}{l} 28 \\ 128 \end{array} \right] \quad \begin{array}{l} p(x=2) = 23.0144 \\ p(x=3) = 35.8528 \end{array} = 23$$

$$P(x=3) = \binom{7}{3} (0.4833)^3 (0.5167)^4 \quad \left[\begin{array}{l} 28 \\ 128 \end{array} \right] \quad \begin{array}{l} p(x=3) = 35.8528 \\ p(x=4) = 33.5104 \end{array} = 36$$

$$P(x=4) = \binom{7}{4} (0.4833)^4 (0.5167)^3 \quad \left[\begin{array}{l} 28 \\ 128 \end{array} \right] \quad \begin{array}{l} p(x=4) = 33.5104 \\ p(x=5) = 18.7904 \end{array} = 34$$

$$P(x=5) = \binom{7}{5} (0.4833)^5 (0.5167)^2 \quad \left[\begin{array}{l} 28 \\ 128 \end{array} \right] \quad \begin{array}{l} p(x=5) = 18.7904 \\ p(x=6) = 5.8496 \end{array} = 19$$

$$P(x=6) = \binom{7}{6} (0.4833)^6 (0.5167)^1 \quad \left[\begin{array}{l} 28 \\ 128 \end{array} \right] \quad \begin{array}{l} p(x=6) = 5.8496 \\ p(x=7) = 0.7808 \end{array} = 6$$

$$P(x=7) = \binom{7}{7} (0.4833)^7 (0.5167)^0 \quad \left[\begin{array}{l} 28 \\ 128 \end{array} \right] \quad \begin{array}{l} p(x=7) = 0.7808 \\ \end{array} = 1$$

we have Recurrence relation for the B.D.

$$P(r+1) = \frac{(n-r)p}{(r+1)q} \cdot P(r).$$

$$\therefore P(1) = \frac{(n-0)p}{(0+1)q} \cdot P(0), \text{ here } \frac{p}{q} = \frac{0.4833}{0.5167} = 0.935$$

$$\rightarrow P(1) = \frac{7}{1} (0.935) (0.0098) = 0.0641$$

$$P(2) = \frac{6}{2} (0.935) (0.0641)$$

$$P(3) = P(2+1) = \frac{5}{3} (0.935) (0.1798)$$

$$P(4) = P(3+1) = \frac{4}{4} \times 0.935 \times 0.2801$$

$$P(5) = P(4+1) = \frac{3}{5} \times 0.935 \times 0.2606$$

Poisson Distribution:- A random variable x follows a poisson distribution with parameter λ if its P.M.F. is $P(X=r) = \frac{e^{-\lambda} \lambda^r}{r!}$, $r=0,1,2,\dots$

→ The P.D. can be derived as a limiting case of the B.D. under the following conditions.

(i), p , the probability of the occurrence of the event is very small.

(ii). n is very very large, where n is no. of trials. i.e. $n \rightarrow \infty$.

(iii) np is a finite quantity, say $np = \lambda$, then λ is called the parameter of the poisson distribution.

Derivation:- $P(r) = \binom{n}{r} p^r q^{n-r} = \binom{n}{r} p^r (1-p)^{n-r}$
 $= \frac{n!}{(n-r)! r!} p^r (1-p)^n = \frac{n(n-1)(n-2)\dots(n-r+1)(n-r)! p^r (1-p)^n}{(n-r)! r! (1-p)^n}$
 $= \frac{n(n-1)(n-2)\dots(n-r+1)}{r!} p^r (1-p)^n$ — (1)

put $np = \lambda \rightarrow n = \frac{\lambda}{p}$

as $n \rightarrow \infty$, $p \rightarrow 0$.

① → $P(r) = \frac{\frac{\lambda}{p}(\frac{\lambda}{p}-1)(\frac{\lambda}{p}-2)\dots(\frac{\lambda}{p}-r+1) p^r (1-p)^n}{r!}$

$= \frac{\lambda(\lambda-p)(\lambda-2p)\dots(\lambda-(r-1)p)}{r!} (1-p)^n$

$= \frac{\lambda(\lambda-p)(\lambda-2p)\dots(\lambda-(r-1)p)}{r! (1-p)^r} (1-\frac{\lambda}{n})^n$ — (2)

applying $\lim_{n \rightarrow \infty}, p \rightarrow 0$ and $np = \lambda$,

$$P(r) = \frac{\lambda^r}{r!} \lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n}\right)^n$$

$$\therefore p(r) = \frac{\lambda^r}{r!} e^{-\lambda}$$

This is known as poisson distribution.

The mean of the Poisson Distribution; —

$$\text{Mean} = \mu = E(x) = \sum_{x=0}^{\infty} x p(x)$$

$$= \sum_{x=0}^{\infty} x \left(\frac{\lambda^x}{x!}\right) e^{-\lambda}$$

$$= \sum_{x=0}^{\infty} x \frac{e^{-\lambda} \lambda^x}{x!}$$

$$= e^{-\lambda} \sum_{x=0}^{\infty} \frac{x \lambda^x}{x(x-1)!} = e^{-\lambda} \sum_{x=1}^{\infty} \frac{\lambda^x}{(x-1)!}$$

$$= e^{-\lambda} \lambda \sum_{x=1}^{\infty} \frac{\lambda^{x-1}}{(x-1)!}$$

$$= e^{-\lambda} \lambda \sum_{y=0}^{\infty} \frac{\lambda^y}{y!}$$

$$= e^{-\lambda} \lambda \cdot e^{\lambda} = \lambda (e^{-\lambda} + \lambda)$$

$$= \lambda (e^0) = \lambda (1) = \lambda$$

$$\begin{aligned} \text{put } x-1 &= y \\ x=1 &\rightarrow y=0 \end{aligned}$$

$$\text{Variance} = \sigma^2 = E(x^2) - (E(x))^2$$

$$= \sum_{x=0}^{\infty} x^2 p(x) - \lambda^2 \quad (\because E(x) = \lambda)$$

$$= \sum_{x=0}^{\infty} x^2 \frac{e^{-\lambda} \lambda^x}{x!} - \lambda^2$$

$$= \sum_{x=0}^{\infty} (x(x-1) + x) \frac{e^{-\lambda} \lambda^x}{x!} - \lambda^2$$

$$= \sum_{x=0}^{\infty} x(x-1) \frac{e^{-\lambda} \lambda^x}{x!} + \sum_{x=0}^{\infty} x \frac{e^{-\lambda} \lambda^x}{x!} - \lambda^2$$

$$= \sum_{x=0}^{\infty} \frac{x(x-1) e^{-\lambda} \lambda^x}{x(x-1)(x-2)!} + \lambda - \lambda^2$$

$$= \sum_{x=2}^{\infty} \frac{e^{-\lambda} \lambda^x}{(x-2)!} + \lambda - \lambda^2 = \lambda^2 e^{-\lambda} \sum_{x=2}^{\infty} \frac{\lambda^{x-2}}{(x-2)!} + \lambda - \lambda^2$$

$$= \lambda^2 e^{-\lambda} \sum_{y=0}^{\infty} \frac{\lambda^y}{y!} + \lambda - \lambda^2 \quad \begin{array}{l} \text{put } x-2=y \\ x=2 \rightarrow y=0 \end{array}$$

$$= \lambda^2 e^{-\lambda} e^{\lambda} + \lambda - \lambda^2$$

$$= \lambda^2 e^{-\lambda + \lambda} + \lambda - \lambda^2$$

$$= \lambda^2 (1) + \lambda - \lambda^2$$

$$= \lambda$$

hence, Mean = Variance = λ .

Recurrence formula:-

$$\text{we have } p(x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$p(x+1) = \frac{\lambda}{x+1} p(x)$$

A manufacturer of cotter pins known that 5% of the product is defective. Pins are sold in boxes of 100. He guarantees that not more than 10 pins will be defective. what is the approximate probability that a box will fail to meet the guaranteed quality?

Soln Given $p = 5\% = \frac{5}{100} = 0.05$.

Total no. of cotter pins = $n = 100$.

mean = $\lambda = np = 100 \times 0.05 = 5$.

$$\text{wkt } p(x=x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$\text{here } \lambda = 5, \text{ so } p(x=x) = \frac{e^{-5} 5^x}{x!}$$

To find the probability that a box will fail to meet the guarantee = $p(x > 10) = 1 - p(x \leq 10)$

$$= 1 - [p(x=0) + p(x=1) + p(x=2) + p(x=3) + \dots + p(x=10)]$$

$$= 1 - \left[\frac{e^{-5} 5^0}{0!} + \frac{e^{-5} 5^1}{1!} + \frac{e^{-5} 5^2}{2!} + \dots + \frac{e^{-5} 5^{10}}{10!} \right]$$

$$= 1 - 0.9863 = 0.0137$$

A car-hire firm has two cars which it hires out day by day. The no. of demands for a car on each day is distributed as a poisson distribution with mean 1.5. calculate the proportion of days (i) on which there is no demand

(ii) on which demand is refused.

Sol

Given mean $= \lambda = 1.5$.

by P.D., we have
$$p(x) = \frac{e^{-\lambda} \lambda^x}{x!}$$
$$= \frac{e^{-1.5} (1.5)^x}{x!}$$

(i) probability of no demand $= p(x=0)$
$$= \frac{e^{-1.5} (1.5)^0}{0!} = e^{-1.5} = 0.2231.$$

No. of days in a year when there is no demand $= 365 \times 0.2231 = 81$ days.

(ii). probability on which demand is refused.

Demand is refused if the no. of demands is more than 2, i.e. $x > 2$.

$$\begin{aligned} \therefore P(x > 2) &= 1 - P(x \leq 2) = 1 - [P(x=0) + P(x=1) + P(x=2)] \\ &= 1 - \left[e^{-1.5} + \frac{e^{-1.5} (1.5)}{1!} + \frac{e^{-1.5} (1.5)^2}{2!} \right] = 1 - e^{-1.5} (1 + 1.5 + 1.125) \\ &= 1 - 3.625 (e^{-1.5}) = 1 - 0.8088 = 0.1913. \end{aligned}$$

& no. of days in a year when same demand is refused $= 365 \times 0.1913 = 69.82 = 70$ days.

Q. If a random variable has a poisson distribution such that $p(1) = p(2)$. Find (i) mean (ii) $p(4)$ (iii) $p(x \geq 1)$ (iv) $p(1 < x < 4)$.

Sol

Given $P(1) = P(2)$

$$\rightarrow \frac{e^{-\lambda} \lambda^1}{1!} = \frac{e^{-\lambda} \lambda^2}{2!}$$

$$\rightarrow 2\lambda = \lambda^2$$

$$\Rightarrow \lambda^2 - 2\lambda = 0$$

$$\rightarrow \lambda(\lambda - 2) = 0 \rightarrow \lambda = 0 \text{ or } \lambda = 2.$$

but $\lambda \neq 0$, $\therefore \lambda = 2$.

(i) $\therefore \text{mean} = \lambda = 2$.

$$(ii) P(4) = \frac{e^{-\lambda} \lambda^4}{4!} = \frac{e^{-2} 2^4}{4!} = 0.09022$$

$$(iii) P(x \geq 1) = 1 - P(x=0) = 1 - \frac{e^{-2} 2^0}{0!} = 1 - \frac{1}{e^2} = 0.8647$$

$$(iv) P(1 < x < 4) = P(x=2) + P(x=3) \\ = \frac{e^{-2} 2^2}{2!} + \frac{e^{-2} 2^3}{3!} = 2e^{-2} + \frac{8}{6}e^{-2} = 0.4511$$

Q. Given that $P(x=2) = 9P(x=4) + 90P(x=6)$ for a poisson variate x . Find (i) $P(x < 2)$ (ii) $P(x > 4)$

(iii) $P(x \geq 1)$.

Ans: $\lambda = 1$.

(i) 0.7358

(ii) 0.0037

(iii) 0.6321

Q. Fit a poisson distribution to the following data

x	0	1	2	3	4	5	Total
$f(x)$	142	156	69	27	5	1	400

Solⁿ

$$\text{mean} = \frac{\sum x_i f(x_i)}{\sum f(x_i)} = 1, N = \sum f(x_i) = 400$$

$\therefore$ Mean of P.D. = $\lambda = 1$.

$$P(X=0) = \frac{e^{-\lambda} \lambda^0}{0!} = \frac{e^{-1} \cdot 1^0}{0!} = e^{-1} \quad (\text{where } P(X) = \frac{e^{-\lambda} \lambda^x}{x!})$$

$$P(X=1) = \frac{e^{-1} \cdot 1^1}{1!} = e^{-1}$$

$$P(X=2) = \frac{e^{-1} \cdot 1^2}{2!} = \frac{e^{-1}}{2}$$

$$P(X=3) = \frac{e^{-1} \cdot 1^3}{3!} = \frac{e^{-1}}{6}$$

$$P(X=4) = \frac{e^{-1} \cdot 1^4}{4!} = \frac{e^{-1}}{24}$$

$$P(X=5) = \frac{e^{-1} \cdot 1^5}{5!} = \frac{e^{-1}}{120}$$

$\therefore$ The required theoretical frequencies

are $400 P(X=x_i)$, $i = 0, 1, 2, 3, 4, 5$.

$400(e^{-1})$, $400(e^{-1})$, $400(\frac{e^{-1}}{2})$, $400(\frac{e^{-1}}{6})$,

$400(\frac{e^{-1}}{24})$, $400(\frac{e^{-1}}{120})$.

$= 147.15$, 147.15 , 73.58 , 24.53 , 6.13 , 1.23 .

$= 147$, 147 , 74 , 25 , 6 , 1 .

hence the Required P.D. is

x	0	1	2	3	4	5
Observed data	142	156	69	27	5	1
Theoretical data	147	147	74	25	6	1

Geometric Distribution:-

The G.D. models the probability that the first success occurs on the k^{th} trial, in a sequence of independent Bernoulli trials.

Each trial has: probability of success = p

probability of failure = $q = 1 - p$

& Trials are independent.

Probability mass function = $p(x=k) = (1-p)^{k-1} p$
where $k = 1, 2, 3, \dots$ & $(1-p)^{k-1}$ = probability of first $k-1$ failures.

Mean and variance:-

$$\text{Mean} = \mu = E(x) = \frac{1}{p}$$

$$\text{Variance} = \text{Var}(x) = \sigma^2 = \frac{1-p}{p^2}$$

$$\text{Standard Deviation} = \sqrt{\frac{1-p}{p^2}} = \frac{\sqrt{1-p}}{p}$$

Q. If probability of successful data transmission is $p = 0.2$, find the probability that first success occurs on 4^{th} attempt.

solⁿ Given $p = 0.2$
probability that first success occurs on

$$4^{\text{th}} \text{ attempt} = p(x=4) = (1-0.2)^3 (0.2)$$

$$= (0.8)^3 (0.2) = 0.512 \times 0.2 = 0.1024$$

Q. The distribution of typing mistakes committed by a typist is given below. Assuming the distribution to be Poisson, find the expected frequencies

x	0	1	2	3	4	5
$f(x)$	42	33	14	6	4	1

Sol $N = \sum f(x_i) = 42 + 33 + 14 + 6 + 4 + 1 = 100$

Mean of the distribution $= \lambda = \frac{\sum x_i f(x_i)}{\sum f(x_i)}$

$\therefore \lambda = \frac{0 \times 42 + 1 \times 33 + 2 \times 14 + 3 \times 6 + 4 \times 4 + 5 \times 1}{100} = \frac{100}{100} = 1$

By poisson distribution $p(x) = \frac{e^{-\lambda} \lambda^x}{x!} = \frac{e^{-1} 1^x}{x!} = \frac{e^{-1}}{x!}$

$\therefore p(0) = \frac{e^{-1}}{0!}$

$p(1) = \frac{e^{-1}}{1!}$

$p(2) = \frac{e^{-1}}{2!}$

$p(3) = \frac{e^{-1}}{3!}$

$p(4) = \frac{e^{-1}}{4!}$

$p(5) = \frac{e^{-1}}{5!}$

and the required

Theoretical frequencies

are $100 p(0), 100 p(1),$

$100 p(2), 100 p(3),$

$100 p(4), 100 p(5)$

$= 100 \times e^{-1}, 100 \times e^{-1},$

$100 \times \frac{e^{-1}}{2}, 100 \times \frac{e^{-1}}{6}, 100 \times \frac{e^{-1}}{24}$

$\frac{100 \times e^{-1}}{120}$

$= 36.79, 36.79, 18.39,$

$6.13, 1.53, 0.31$

& the nearest integers are 37, 37, 18, 6, 2, 0.

here we get

x	0	1	2	3	4	5
Observed data	42	33	14	6	4	1
Expected data	37	37	18	6	2	0